Reliability Calculator (IEC TR 62380)

[Z] Reliability Calculator rev_motherboard

Project: /home/eliot/Documents/Xplore/Avionics_HW/node_pcb/AV_motherboard/AV_pcb/rev_motherboard

Open... | Save | Analysis Suite | Export Report

Schematic Sheets

- /Motherboard/Ethernet/
- /Motherboard/Local Power/
- /Motherboard/USBc port/
- /Motherboard/Interfacing/
- /Motherboard/Flags/
- /Motherboard/Sensors/
- /Motherboard/RS485/
- /Motherboard/EEPROM/
- /Motherboard/Power - Power Path/
- /Motherboard/MCU1/
- /Motherboard/Hat_connections/Hat_connector_2/

Add Selected | Add All

Edit Components...

Mission Profile

Template: LEO Satellite

Duration	3	years
Thermal cycles	5256	/year
dT per cycle	3.0	degC
Default tau_on	1.00	(0-1)

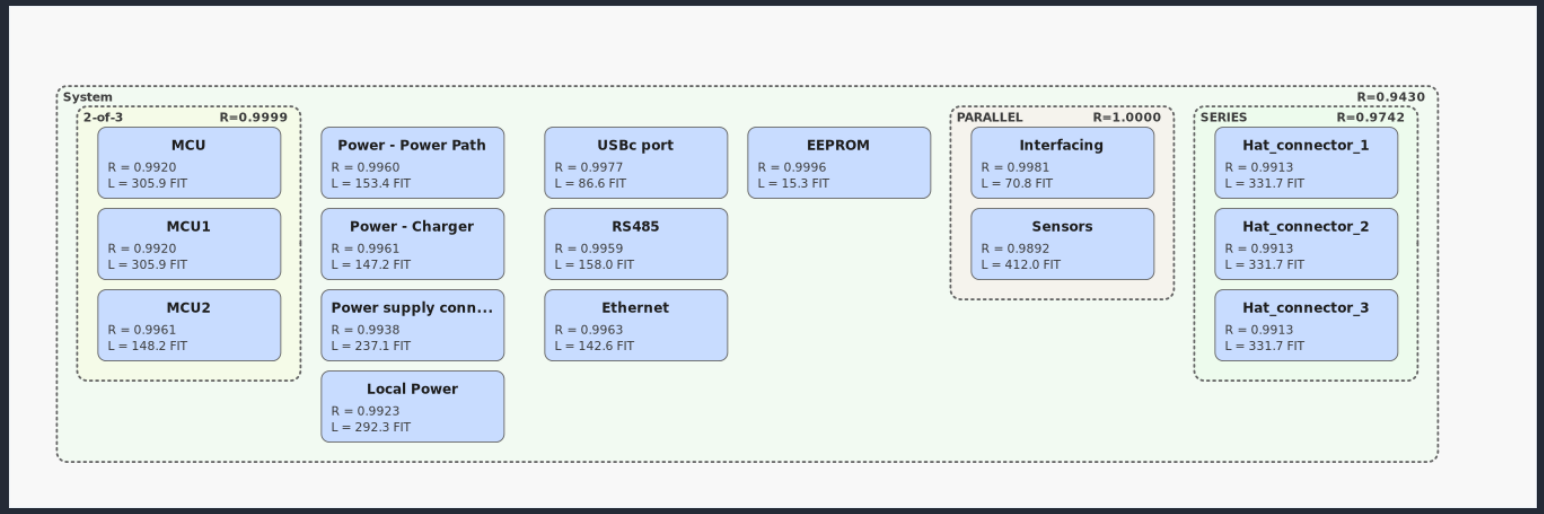
Phases: (3 defined)

Phase	Dur%	T_amb	T_junc	Cycles	dT	tau
Sunlit	60	60	60	5256	30.0	1.00
Eclipse	37	-20	-20	5256	30.0	0.80
Safe Mode	3	0	0	100	5.0	0.10

+ Phase | Edit | - Phase

IEC 62380 Reference...

System Block Diagram



Components -- Hat_connector_3

Edit

Ref	Value	Type	L (FIT)	R
R30	10k	Resistor	0.1	1.00000
JP7	JP_I2C_5V_3V3	Connector	35.4	0.99907
D7	RCLAMP3346..	Diode	4.4	0.99989
D5	RCLAMP3346..	Diode	4.4	0.99989
TP2	PG_3V3	Miscellaneous	52.6	0.99862
JP5	JP_GPIO_5V_3...	Connector	35.4	0.99907
J5	Conn_PicoBl..	Connector	35.4	0.99907
D6	RCLAMP3346..	Diode	4.4	0.99989
JP4	JP_SPI_5V_3V3	Connector	35.4	0.99907
J3	Conn_PicoBl..	Connector	35.4	0.99907
R29	10k	Resistor	0.1	1.00000

Sheet: L = 384.3 FIT R = 0.98995

System Results

```
==========================================
      SYSTEM RELIABILITY ANALYSIS
==========================================

 Mission: 3.0 years (26280 h)

  Reliability:  R = 0.943001
  Failure Rate: L = 2233.19 FIT
  MTTF: 51.1 years

==========================================
           BLOCK DETAILS
==========================================

 Ethernet
   L = 142.6 FIT, R = 0.996259

 MCU1
```

Recalculate

System R = 0.943001 (3.0y)

Close

Reliability Calculator (IEC TR 62380)

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Project: /home/eliot/Documents/Xplore/Avionics_HW

Analysis Suite | Export Report

Schematic Sheets

/Motherboard/Ethernet/
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/Motherboard/RS485/
/Motherboard/EEPROM/
/Motherboard/Power - Power Path/
/Motherboard/MCU1/
/Motherboard/Hat connections/Hat_connector_2/

Add Selected

Edit Components...

Mission Profile

Template: LEO Satellite

Duration 3

Thermal cycles 5256

dT per cycle 3.0

Default tau_on 1.00

Phases: (3 defined)

Phase	Dur%	T_amb	T_junc	Cycles
Sunlit	60	60	60	5256
Eclipse	37	-20	-20	5256
Safe Mode	3	0	0	100

+ Phase | Edit | - Phase

IEC 62380 Reference...

Batch Component Editor

Components:
Auto-classify status: 52/75 high-confidence, 16 need review, 7 manual.

Ref	Value	Type	Lambda (FIT)	Status	Why
R715	40k 0.1%	Resistor	0.1	High	Reference prefi...
R704	0R 1%	Resistor	0.1	High	Reference prefi...
C712	470n X7R 50V	Capacitor	0.8	High	Reference prefi...
Q704	BSC059N04LS..	Transistor	5.1	High	Reference prefi...
TP718	GND	Miscellaneous	52.6	Review	Reference or fo...
R703	0R 1%	Resistor	0.1	High	Reference prefi...
R709	0R 1%	Resistor	0.1	High	Reference prefi...
TP714	REGN	Miscellaneous	52.6	Review	Reference or fo...
TP715	BTST1	Miscellaneous	52.6	Review	Reference or fo...
C707	56u 63V	Capacitor	0.8	High	Reference prefi...
TP712	CEn	Miscellaneous	52.6	Review	Reference or fo...
TP717	+3V3	Miscellaneous	52.6	Review	Reference or fo...
C711	100n X7R 50V	Capacitor	0.8	High	Reference prefi...
C717	1u X7R 50V	Capacitor	0.8	High	Reference prefi...
R702	5m 1% 1W	Resistor	0.1	High	Reference prefi...
R705	10R 1%	Resistor	0.1	High	Reference prefi...
R701	5m 1% 1W	Resistor	0.1	High	Reference prefi...
NT704	NetTie_OUT-	Miscellaneous	0.0	Manual	User reviewed ...
NT701	NetTie_IN+	Miscellaneous	0.0	Manual	User reviewed ...
R708	10R 1%	Resistor	0.1	High	Reference prefi...
C713	100n X7R 50V	Capacitor	0.8	High	Reference prefi...
C718	100n X7R 50V	Capacitor	0.8	High	Reference prefi...
C701	56u 63V	Capacitor	0.8	High	Reference prefi...
C709	470n X7R 50V	Capacitor	0.8	High	Reference prefi...
R710	0R 1%	Resistor	0.1	High	Reference prefi...

Edit Selected... | Auto-Classify All | Next Flagged

Quick Edit:

Type: Integrated Circuit

Lambda Override

☐ Override with fixed FIT value

Lambda: 0.000 FIT

Required Fields

Ic Type *
IC technology and function (Table 16)
MOS Digital (Micro/DSP)

Transistor Count *
Count in type's native unit (gates, bits, macrocells, transistors - see Table 16)
10000

T Junction *
Junction temperature (C)
85.000

Package *
Package type (Table 17a/17b)
QFP-48 (7x7mm)

Optional / Environmental

Construction Year
Fabrication year (technology maturity, a = year - 1998)
2020

Substrate

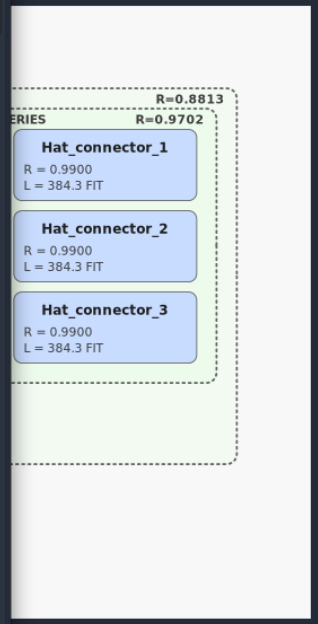
Status: Review | Source: mechanical heuristic | Reference or footprint looks mechanical/test-only

Apply to Selected

Cancel | Save All

NT703	NetTie_OUT+	Miscellaneous	52.6	0.99862
R717	1k 1%	Resistor	0.1	1.00000

Sheet: L = 1356.7 FIT R = 0.96497



Recalculate

Saved

Close

System Reliability Analysis

Reliability Analysis Suite

λ = 2233.19 FIT | R(t) = 0.943001 | 3.0y | 16 sheets

Clear All | Cancel

Overview | Analysis | Design Actions | Report

Design Overview

This page highlights current reliability risk, the biggest contributors, and the next analysis to run.

Component Types (uncheck to exclude from all analyses):

- [x] Capacitor
- [x] Connector
- [x] Crystal/Oscillator
- [x] Diode
- [x] Inductor/Transformer
- [x] Integrated Circuit
- [x] Miscellaneous
- [x] Resistor
- [x] Transistor

System FIT	Mission R(t)	MTTF	Components	Classification Review
2353.02 FIT	0.940036	48.5 years	382 across 16 sheets	324 high-confidence, 3 flagged

Next Best Action

Review 3 flagged component classifications before publishing a report.

What This Means

The estimate is usable, but some component categories are still heuristic and should be confirmed.

FIT Contributions (Top 15)

System Summary

```
System λ:    2353.02 FIT
R(t):         0.940036
MTTF:         48.5 years
Mission:      3.0 years
Components:   382 (3 flagged for review)
Sheets:       16
```

Component	Type	λ (FIT)	Contribution %	Cumulative %
Y3	Miscellaneous	52.59	2.2%	2.2%
FB3	Miscellaneous	52.59	2.2%	4.5%
Y5	Crystal/Oscillator	52.59	2.2%	6.7%
Y4	Crystal/Oscillator	52.59	2.2%	8.9%
SW4	Miscellaneous	52.59	2.2%	11.2%
SW3	Miscellaneous	52.59	2.2%	13.4%
J8	Connector	35.39	1.5%	14.9%
J23	Connector	35.39	1.5%	16.4%
J21	Connector	35.39	1.5%	17.9%

Ready

System Reliability Analysis

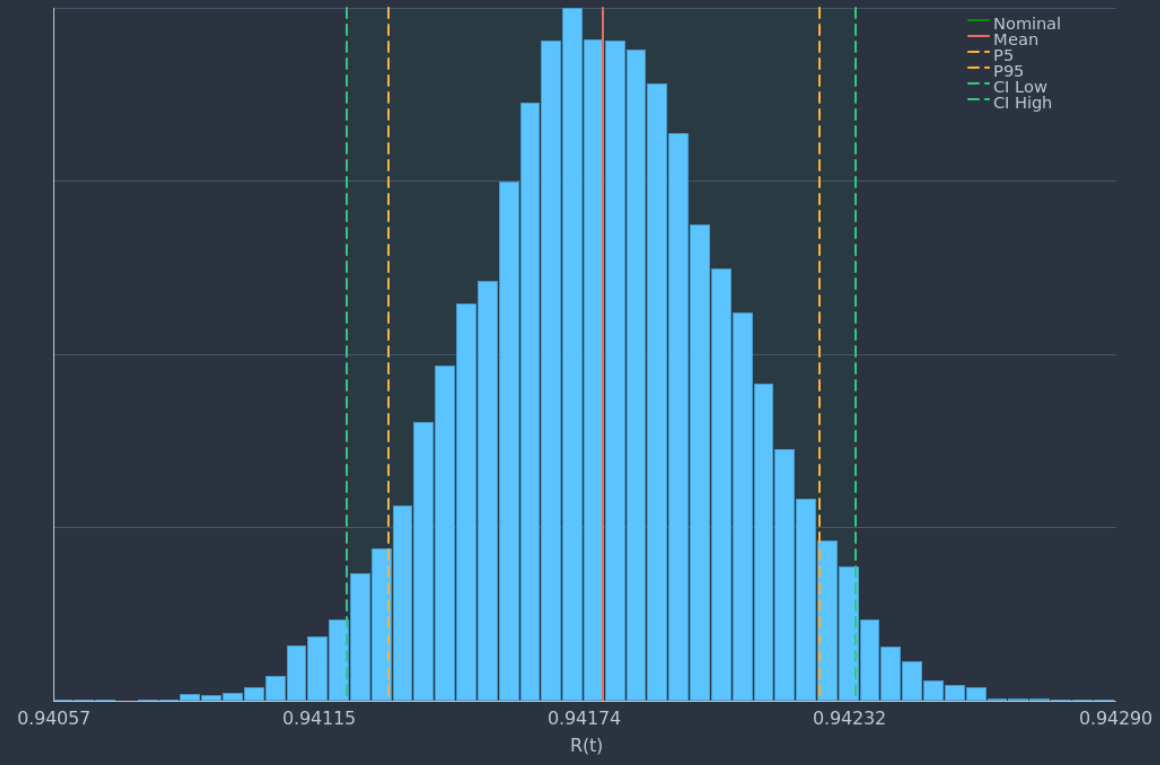
Reliability Analysis Suite

λ = 2233.19 FIT | R(t) = 0.943001 | 3.0y | 16 sheets

Clear All | Cancel

Overview | Analysis | Design Actions | Report

R(t) Distribution



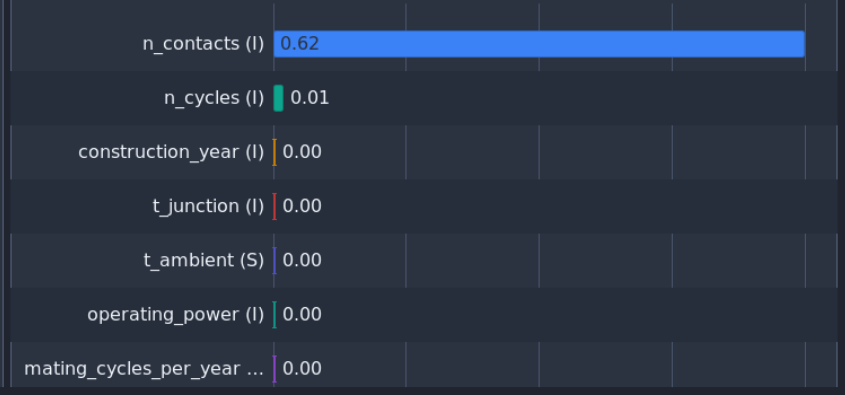
Statistics

```
Nominal R(t):   0.940036
Mean R(t):      0.941779
Median R(t):    0.941778
Std Dev:        0.000286
CI [95%]:      [0.941218, 0.942335]
CI Width:       0.001117
Mean HW CI:     ±5.61e-06

Mean λ:        2282.52 FIT
Std λ:         11.56 FIT
CI λ:          [2260.06, 2305.20]

Simulations:    10,000
Runtime:        27.6s
Uncertain params: 18
Uncertain comps:  330/382
```

Parameter Importance (SRRC²)



Uncertainty analysis complete: 10000 samples in 27.6s

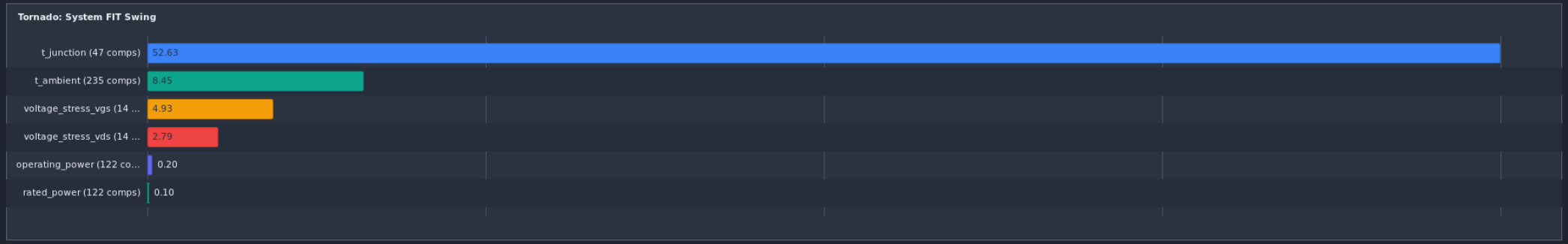
System Reliability Analysis

Reliability Analysis Suite

λ = 2233.19 FIT | R(t) = 0.943001 | 3.0y | 16 sheets

Clear All | Cancel

Overview | Analysis | Design Actions | Report



Parameter	Low FIT	Base FIT	High FIT	Swing	Perturbation
t_junction (47 comps)	2329.64	2353.02	2382.28	52.63	-10.0 degC / +10.0 degC
t_ambient (235 comps)	2348.94	2353.02	2357.39	8.45	-10.0 degC / +10.0 degC
voltage_stress_vgs (14 comps)	2350.74	2353.02	2355.67	4.93	-0.05 ratio / +0.05 ratio
voltage_stress_vds (14 comps)	2351.68	2353.02	2354.47	2.79	-0.05 ratio / +0.05 ratio
operating_power (122 comps)	2352.92	2353.02	2353.12	0.20	-0.01 W / +0.01 W
rated_power (122 comps)	2353.09	2353.02	2352.99	0.10	-0.05 W / +0.05 W

Interpretation: prioritize work on 't_junction (47 comps)' first; it changes system FIT more than the other currently tested perturbations.

Step 3: Component Criticality (Elasticity)

Elasticity measures how sensitive the system failure rate is to each component's parameters. A 10% perturbation is applied to each numeric field; the resulting change in component FIT is normalized to give a dimensionless elasticity: e = (ΔΛ/Λ) / (Δx/x). Higher elasticity = more design leverage. 'Impact %' shows the contribution of that parameter's swing relative to the total system FIT.

Top N (0 = all): 0 –

▶ Run Criticality

Reference	Type	λ (FIT)	Top Parameter	Elasticity	Impact %
Y3	Miscellaneous	52.59	n_cycles	0.616	12.3%
FB3	Miscellaneous	52.59	n_cycles	0.616	12.3%
Y5	Crystal/Oscillator	52.59	n_cycles	0.616	12.3%
Y4	Crystal/Oscillator	52.59	n_cycles	0.616	12.3%
SW4	Miscellaneous	52.59	n_cycles	0.616	12.3%
SW3	Miscellaneous	52.59	n_cycles	0.616	12.3%
J8	Connector	35.39	n_contacts	0.972	19.4%
J23	Connector	35.39	n_contacts	0.972	19.4%
J21	Connector	35.39	n_contacts	0.972	19.4%

Criticality: 330 components analyzed

System Reliability Analysis

Reliability Analysis Suite

λ = 2233.19 FIT | R(t) = 0.943001 | 3.0y | 16 sheets

Clear All | Cancel

Overview | Analysis | Design Actions | Report

Smart Design Actions (auto-identified from analyses)

▶ Identify Best Actions

Combines Tornado + SRRC + Criticality to find the parameters with the highest design leverage. Run Step 2/3 first for best results.

#	Parameter	Score	FIT Gain	Suggestion	Components	Source
1	n_contacts	0.986	0.0	Reduce n_contacts to lower system FIT by up to 0.0%	J8, J23, J21, J22, J24 (+5)	SRRC + Criticality
2	delta_t	0.628	0.0	Reduce delta_t to lower system FIT by up to 0.0%	Y3, FB3, Y5, Y4, SW4 (+5)	Criticality
3	t_junction	0.563	26.3	Reduce junction temperature via better packages or heatsinking	U12, U10, U11, U17, U16 (+5)	Tornado + SRRC + Criticality
4	n_cycles	0.361	0.0	Reduce n_cycles to lower system FIT by up to 0.0%	Y3, FB3, Y5, Y4, SW4 (+5)	SRRC + Criticality
5	voltage_stress_vgs	0.150	2.5	Reduce voltage stress ratio (use higher-rated components)	Q3, Q2, Q4, Q801, Q804 (+5)	Tornado + SRRC + Criticality
6	tau_on	0.148	0.0	Reduce tau_on to lower system FIT by up to 0.0%	Y3, FB3, Y5, Y4, SW4 (+5)	Criticality
7	power_loss	0.124	0.0	Derate operating power (use components at lower % of rated power)	L2, L1, LV2, LV1, LED801 (+4)	SRRC + Criticality
8	voltage_stress_vds	0.085	1.4	Reduce voltage stress ratio (use higher-rated components)	Q3, Q2, Q4, Q801, Q804 (+5)	Tornado + SRRC + Criticality
9	t_ambient	0.082	4.2	Improve thermal design to lower ambient temperature	L2, L1, LV2, LV1, LED801 (+5)	Tornado + SRRC + Criticality

Top action: n_contacts -- This parameter has a 0.0% impact on system FIT based on OAT sensitivity analysis.

1. What-If / Design Margin Scenarios

▶ Run Scenarios

Evaluates design-actionable and environmental scenarios. Each recomputes every component through IEC TR 62380.

Custom: Name | construction_year | multiply | Value | + Add

Scenario	Description	λ (FIT)	R(t)	Δλ %	ΔR
Derate power 20%	Reduce operating power by 20% on all components (better thermal design)	2353.00	0.940036	-0.0%	+0.000000
Derate power 50%	Reduce operating power by 50% (aggressive derating for space)	2352.97	0.940037	-0.0%	+0.000001
Better packages (-15C Tj)	Upgrade packages to reduce junction temperature by 15 degC	2319.87	0.940855	-1.4%	+0.000819
Better packages (-30C Tj)	Premium thermal packages: junction temperature reduced 30 degC	2296.94	0.941422	-2.4%	+0.001386
Hi-rel parts (-30% base FIT)	Use mil-spec / space-grade parts with 30% lower base failure rates	1647.11	0.957637	-30.0%	+0.017602
Hi-rel parts (-50% base FIT)	Use highest-reliability screened parts with 50% lower base failure rates	1176.51	0.969554	-50.0%	+0.029519
Voltage derating 20%	Reduce all voltage stress ratios by 20% (conservative design)	2346.87	0.940188	-0.3%	+0.000152
Voltage derating 50%	Aggressive voltage derating for high-reliability (50% reduction)	2341.69	0.940316	-0.5%	+0.000280
Temp +10 degC	Environment: all temperatures +10 degC	2386.65	0.939205	+1.4%	-0.000830
Temp -10 degC	Improved cooling: all temperatures -10 degC	2325.57	0.940714	-1.2%	+0.000678

2. Reliability Target Closure Planning

R target: 0.999 | Strategy: Proportional (Recommended) | Margin %: 10 | ▶ Allocate Budget

Scenarios: 10 evaluated

System Reliability Analysis

Reliability Analysis Suite

λ = 2233.19 FIT | R(t) = 0.943001 | 3.0y | 16 sheets

Clear All | Cancel

Overview | Analysis | Design Actions | Report

1. What-If / Design Margin Scenarios

▶ Run Scenarios

Evaluates design-actionable and environmental scenarios. Each recomputes every component through IEC TR 62380.

Custom: Name | construction_year | multiply | Value | + Add

Scenario	Description	λ (FIT)	R(t)	Δλ %	ΔR
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Temp +10 degC	Environment: all temperatures +10 degC	2386.65	0.939205	+1.4%	-0.000830
Temp -10 degC	Improved cooling: all temperatures -10 degC	2325.57	0.940714	-1.2%	+0.000678

2. Reliability Target Closure Planning

R target: 0.9 | Strategy: Complexity | Margin %: 20 | ▶ Allocate Budget

How it works: Converts your target R into a total FIT ceiling, applies the requested design margin, and then shows the remaining gap to close. The strategy changes how the allowable FIT is apportioned, but the main outputs are always the same: total gap, pressured sheets, and the components that must save the most FIT.
Strategy note: Proportional preserves the current design balance and is the recommended baseline. Equal, Complexity, and Criticality are alternate lenses for planning conversations.

✅ Target R=0.9 gives 4009.2 FIT before margin and 3207.3 FIT after margin. Actual = 2353.0 FIT. No closure work is required. Strategy: Complexity (weights heavier sheets more strongly).

Effective Budget
3207.3 FIT (20% margin)

Actual FIT
2353.0 FIT

Gap To Close
Within budget

Sheet	Actual FIT	Budget FIT	Gap To Close	Util %	Hotspots
Ethernet	142.60	344.24	0.00	41%	5
Local Power	292.26	377.83	0.00	77%	8
USBc port	86.61	117.55	0.00	74%	4
Interfacing	70.78	16.79	53.99	422%	2
Sensors	412.03	209.90	202.13	196%	11

Reference	Type	Actual FIT	Budget FIT	Save Needed	Util %	Status	Why

Budget: 382 components allocated

Reliability Analysis Report

Project: **rev_motherboard** | 2026-04-10 13:43 | IEC TR 62380:2004

Prepared by **Eliot Abramo**

Contents: What This Report Means · System Summary · Executive Summary · Methodology · Monte Carlo Uncertainty · Tornado Sensitivity · What-If Scenarios · Derating Guidance · Component Criticality · Budget Allocation · Swap Analysis · Glossary · FIT Contributions (Annex) · Sheet Details (Annex)

What This Tool Does And What This Report Communicates

This plugin turns your KiCad design into a reliability model. It reads the component set, applies the IEC TR 62380 failure-rate model to each part, combines those part-level rates into a system result, and then adds uncertainty and sensitivity views so you can see what is solid, what is assumption-driven, and what is worth fixing first.

In plain terms, this report tells you four things. First, the current estimate: system FIT, mission reliability, and MTTF. Second, how much that estimate moves when uncertain inputs such as temperature, duty, or cycling are allowed to vary. Third, which sheets, components, and parameters are doing most of the damage. Fourth, where design effort is most likely to buy back margin if the result is not where you want it yet.

It is best used as a design-review document. The summary gives the headline result, the analysis pages explain what is driving it, and the action pages help you decide what to change next. For the math, assumptions, and interpretation limits behind the numbers, see `docs/METHODOLOGY.md`.

Component classification provenance: 282 high-confidence auto-classified, 0 explicit from fields, 1 manually reviewed, 0 flagged for review.

System Summary

0.943001	2233.2	51.1	3.0
System Reliability	Failure Rate (FIT)	MTTF (years)	Mission (years)

Excellent Mission: 5256 cycles/yr, dT = 3.0 degC